



6. Eigenspaces

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We now know that the eigenvalues of a matrix are the roots of the characteristic equation $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$. Let us proceed to find the eigenvectors corresponding to each eigenvalue.

Definition 6.1

The **eigenspace** of an eigenvalue λ of a square matrix $\mathbf{A}$ is the set of all eigenvectors corresponding to λ , together with the zero vector $\mathbf{0}$. In otherwords, it is the span of all the eigenvectors associated to the eigenvalue λ .

In other words,

the eigenspace of an eigenvalue λ , $= \{\text{span of all eigenvectors associated to } \lambda\}$

$= \{\text{all solutions to } (\lambda \mathbf{I} - \mathbf{A}) \mathbf{v} = \mathbf{0}\}$

$= \text{NS}(\lambda \mathbf{I} - \mathbf{A}).$

There is one eigenspace for every eigenvalue. Each eigenspace is a vector space, so it can be described as the span of a basis. To compute the eigenspace of λ , compute $\text{NS}(\lambda \mathbf{I} - \mathbf{A})$ by Gaussian elimination and back-substitution.

Finding eigenvalues and eigenvectors: a 2 by 2 example



There's a whole line of eigenvector, and I just want a basis. And for a line I just want one vector. But you could-- there's some freedom in choosing that one, but choose a reasonable one. What's a vector in the null space of that? Well the natural vector to pick as the eigenvector with lambda 2 is minus 1, 1. If I did elimination on that vector and set that-- the free variable-- to be 1, I would get minus 1, and get that eigenvector. So do you see then that I've got eigenvector, eigenvalue, eigenvector, eigenvalue for this matrix?

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Steps to find all eigenvectors associated to a given eigenvalue λ of a square matrix $\mathbf{A}$:

1. Write down $\lambda \mathbf{I} - \mathbf{A}$.
2. Use Gaussian elimination to find a basis of $\text{NS}(\lambda \mathbf{I} - \mathbf{A})$.
3. The eigenvectors corresponding to the eigenvalue λ are all the linear combinations of these basis vectors (but not including the zero vector).

Example 6.2 Find all the eigenvalues, eigenvectors, and eigenspaces of $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

Solution:

The eigenvalues are **0**, **1**, and **2**. (We have found these before, by determining the roots of the $P(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A})$).

Eigenspace of **0**:

This is $\text{NS}(\mathbf{0I} - \mathbf{A}) = \text{NS}(-\mathbf{A}) = \text{NS}(\mathbf{A})$, i.e. the set of all solutions to

$$\mathbf{Av} = \mathbf{0}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

The first and third rows are the same, and so subtracting row 1 from row 3, we get

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The resulting matrix is in reduced-row-echelon form. The null space of $\mathbf{A}$ is

$$\text{Eigenspace of } 0 = \text{NS}(\mathbf{A}) = \text{Span} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

In other words, the eigenspace of the eigenvalue 0 is the 1-dimensional vector space spanned by the eigenvector $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, and the set of all eigenvectors is all nonzero scalar multiples of $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$. Geometrically, the eigenspace consists of all vectors along the line containing $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, which is defined by $x = -z$, $y = 0$.

Terminology: Very often, we say “the eigenvectors of λ ” to mean a **basis** of the eigenspace of λ even though strictly speaking, the eigenvectors are all nonzero combinations of that basis.

Eigenspace of 1:

This is $\text{NS}(\mathbf{I} - \mathbf{A})$, i.e. the set of all solutions to

$$(\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$$

$$\begin{pmatrix} 1-1 & 0 & -1 \\ 0 & 1-1 & 0 \\ -1 & 0 & 1-1 \end{pmatrix} \mathbf{v} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

We can immediately see that the first and third components of $\mathbf{v}$ must be zero, while the second component is a free parameter. Hence, the set of all solutions to $(\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$ is

$$\text{Eigenspace of } 1 = \text{NS}(\mathbf{I} - \mathbf{A}) = \text{Span} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

The eigenspace of the eigenvalue 1 is the 1-dimensional vector space spanned by the eigenvector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.

Geometrically, this consists of all vectors along the y -axis.

Eigenspace of 2:

This is $\text{NS}(\mathbf{2I} - \mathbf{A})$, i.e. the set of all solutions to

$$(\mathbf{2I} - \mathbf{A})\mathbf{v} = \mathbf{0}$$

$$\begin{pmatrix} 2-1 & 0 & -1 \\ 0 & 2-1 & 0 \\ -1 & 0 & 2-1 \end{pmatrix} \mathbf{v} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

The matrix can be reduced to:

